

CSCE 670 · DATA MINING & ANALYSIS · FINAL PROJECT

Instacart Market Basket Analysis: Sequential & Associative Pattern Mining

Combining FP-Growth and PrefixSpan to uncover the hidden temporal rhythm of grocery shopping behavior across 3.4 million orders.

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Texas A&M University · Spring 2026

The Grocery Blind Spot

Why today's recommendation engines are fundamentally incomplete

Single-Basket Thinking: Current systems only analyze what's in today's cart.

FP-Growth finds co-purchases but ignores the order of purchases over time.

No Temporal Memory: They ignore the purchasing timeline of individual users.

Flour on Monday → eggs next week — no current system captures this.

The Cost: Wasted perishable inventory and millions in missed revenue.

Our EDA confirmed >99.9% sparsity — most items in <1% of baskets.

3.4M

ORDERS

200K+

USERS

50K+

PRODUCTS



Predicting the Future Cart

Two complementary mining approaches on real Instacart data

Real-World Scale: 778K+ transactions across 5,000 sampled users.

Reproducible sampling (seed=42) from the 3.4M order Instacart dataset.

Two-Pronged Approach: FP-Growth for co-occurrence + PrefixSpan for sequences.

Course technique: "bought together" · External: "bought next".

Basket Trajectories: Chronological sequence of canonicalized basket tuples.

Proper nested structure preserving order boundaries.

WEEK 1

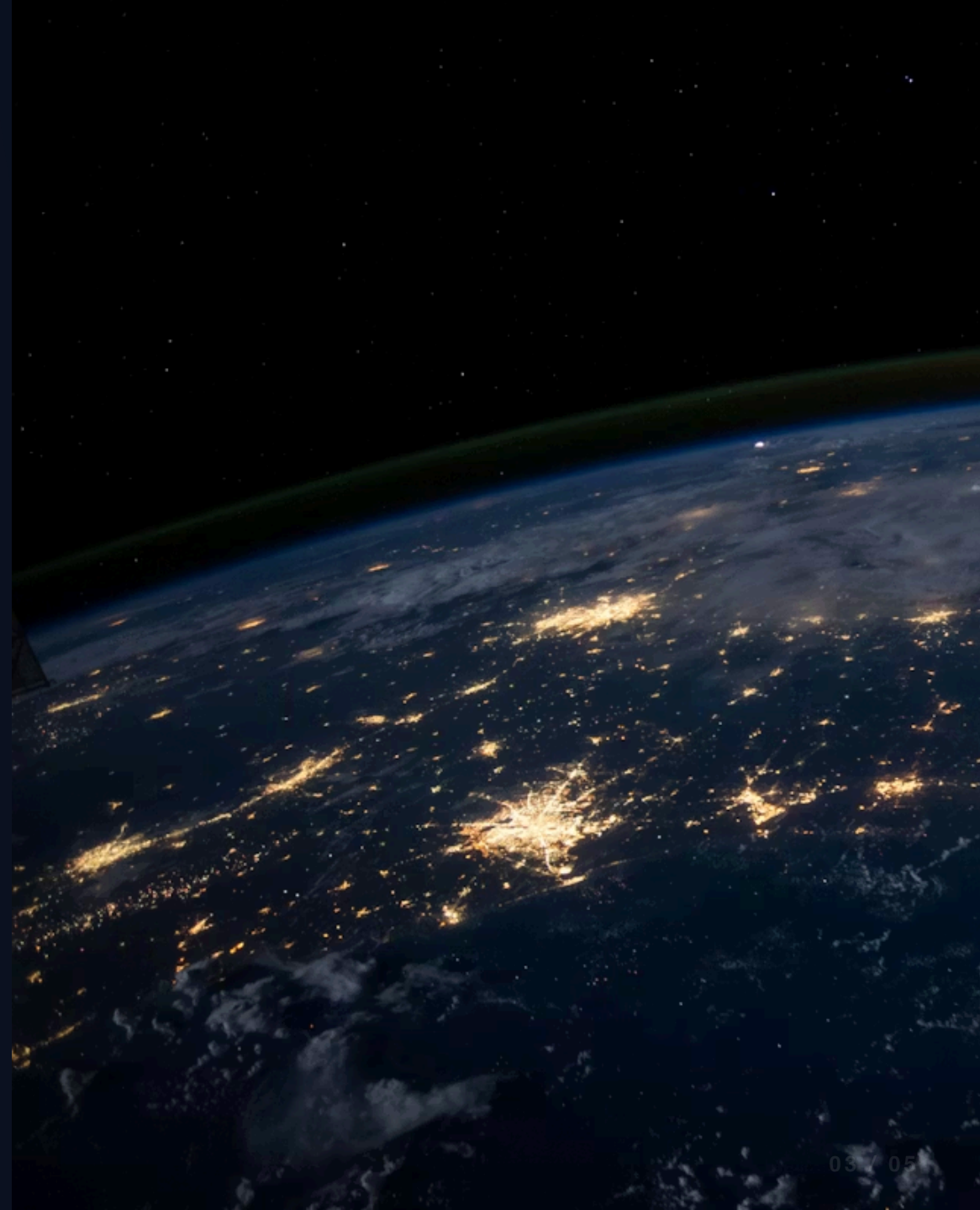
🍌 Banana, 🥛 Milk

WEEK 2

🍌 Banana, 🥚 Eggs

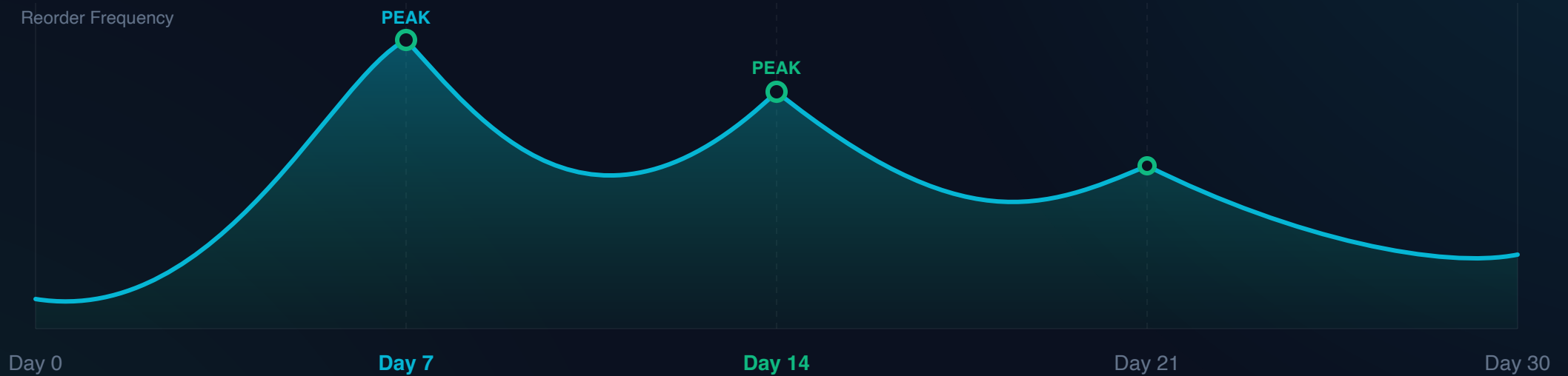
WEEK 3

🍌 Banana, 🍓 Berries



The 7 & 14-Day Rhythm

EDA on 5,000 sampled users reveals strict, predictable shopping cycles



Distinct reorder spikes at **7, 14, 21, and 30 day** intervals — strict weekly rhythms.

Over **99.9% sparsity** — most of 50K products appear in <1% of baskets.

Early-week vs late-week shoppers show **distinct** basket compositions.

What We Found & Where It Goes

Concrete results from 778K+ real transactions

344 **Association Rules** via FP-Growth at 0.5% support across 56K baskets

10K+ **Sequential Patterns** mined by PrefixSpan across basket trajectories

3+ **Multi-Step Trajectories** — basket-to-basket-to-basket sequences only
PrefixSpan can capture

3 **Research Questions** fully answered: thresholds, segments, sequences

The Takeaway: PrefixSpan captures multi-step purchasing trajectories that co-occurrence analysis structurally cannot represent — the ordered sequence of *what comes after what*.

This trajectory structure enables next-basket prediction and demand forecasting beyond what static basket analysis can provide.

The Opportunity: Pre-stock warehouses using 7-day cycle forecasting, push perfectly timed coupons, build time-aware recommendations.

Transform millions of grocery receipts into a roadmap for predictive behavioral logistics.